

Business & Operating Model

Reference framework for strategy, operations, systems, data, and analytics

Purpose

Northstar IT is a synthetic B2B technology company designed as a realistic environment for business planning, process design, relational database development, SQL analysis, financial reporting, and operational decision support.

Working business definition

Northstar IT sells Northstar ServiceOps, a SaaS service-management and operations platform for small and midsize IT service providers. Recurring software revenue is supported by implementation, data migration, training, consulting, and integration services.

110	225	196	\$7.9M
unique employees in HR source	customer accounts	active customers	current ARR

This document intentionally defines only the operating model needed to support Northstar's database, analytics, and portfolio case study. It is not intended to simulate every document a 110-person company would maintain.

01 EXECUTIVE SNAPSHOT

How the business fits together

Northstar's operating logic is simple: acquire the right customers, implement them successfully, drive product adoption, support them reliably, retain and expand recurring revenue, and use integrated people/customer/financial data to make better operating decisions.



Strategic posture

- Growth-stage company: recurring revenue is expanding while the organization works toward stronger operating leverage.
- B2B SaaS + professional services: recurring subscriptions are the core model; services support successful adoption and complex customer needs.
- Operationally data-driven: customer, workforce, service, support, billing, and finance data are designed to connect rather than live in separate silos.
- Management challenge: growth must be balanced with implementation capacity, support quality, product adoption, retention, and financial discipline.

Core management questions

- Are we growing recurring revenue at a rate the delivery organization can support?
- Which customers, plans, and services create the strongest long-term value?
- Where are implementation delays, support demand, or low adoption increasing churn risk?
- Are staffing levels and labor costs aligned with customer demand and service commitments?
- Where are budget variances signaling a process, capacity, pricing, or execution problem?
- What should leadership change now to improve margin, retention, service quality, and scalability?

02 BUSINESS MODEL

What Northstar sells and how it makes money

Primary offering

Northstar ServiceOps is a B2B SaaS platform for small and midsize IT service providers. The product supports service-management and operational workflows, with plan tiers that increase automation, reporting, workflow capability, and multi-team complexity.

Plan	Monthly Base	Included Seats	Extra Seat	Positioning
Starter	\$899	10	\$79	Small IT providers
Professional	\$2,499	30	\$69	Growing teams / more automation
Scale	\$5,499	75	\$59	Multi-team / multi-location

Additional revenue streams

Add-ons: Advanced Analytics, Asset Management, Security & Compliance, API & Integrations, Single Sign-On.

Professional services: Implementation, Data Migration, Training, Consulting.

Customer profile

The synthetic customer base contains 225 historical accounts, with 196 active as of the current model date. Customers are segmented as Small, Lower Mid-Market, and Mid-Market organizations across several industries. The operating model assumes Northstar competes primarily on workflow fit, implementation quality, service reliability, reporting, and product adoption.

03 ORGANIZATION & ACCOUNTABILITY

How the 110-person workforce supports the model

Operating structure

Function	Headcount	Primary accountability
Executive	4	Strategy, operating priorities, technology direction, finance, people leadership
Sales & Account Management	16	Pipeline, new business, account ownership, expansion, commercial relationship
Service Desk	33	Ticket intake, triage, SLA management, resolution, escalation, customer support
Field Services	15	Implementation/deployment work, on-site or project delivery
Network & Infrastructure	14	Technical delivery, integrations, infrastructure, complex escalation
Software & Data	14	Product development, data, reporting, internal technical capability
Finance	7	Billing, receivables, budgeting, financial reporting, management analysis
People & Administration	7	HR, workforce administration, policy, recruiting, training support, administration

Design principle

The org chart should clarify reporting relationships, but the operating model should clarify cross-functional ownership. Customer outcomes span Sales, Implementation, Support, Product, Finance, and leadership; no single department can optimize the lifecycle alone.

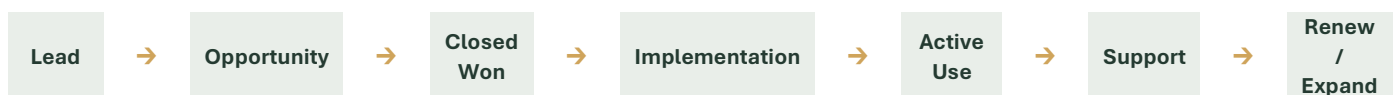
High-level responsibility model

Process	Primary Owner	Key Partners	Key Control	Output
Sales pipeline	Sales	Executive / Finance	Stage discipline + pricing guardrails	Qualified bookings
Implementation	Field / Technical Delivery	Sales / Customer Success	Scope, target go-live, hours	Successful go-live
Customer support	Service Desk	Technical teams / Customer Success	Priority, SLA, escalation	Resolved issue + CSAT
Renewal / expansion	Account Management	Support / Product / Finance	Adoption + service health + commercial review	Retained / expanded ARR
Monthly reporting	Finance / Operations	All functional leaders	Data-quality and close checklist	P&L, BvA, KPIs
Workforce planning	People / Operations	Finance / functional leaders	Capacity + skills + cost review	Hiring / development decisions

04 CORE OPERATING PROCESSES

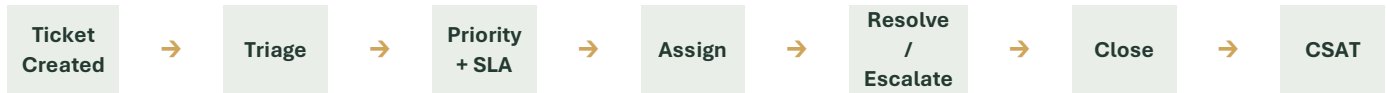
The workflows the database must support

Customer lifecycle



Data should preserve commercial history, implementation outcomes, product usage, support experience, billing, and renewal/churn events.

Support lifecycle



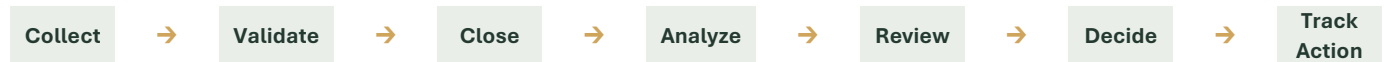
The support dataset must allow analysis of volume, priority, resolution time, SLA performance, escalation, agent workload, and satisfaction.

Employee lifecycle



HR source data already represents recruiting, employee master data, job/compensation history, training, attendance, PTO, and exits.

Management reporting cycle



Reporting is not the end product. The purpose of the data model is to support repeatable management decisions and follow-through.

05 SYSTEMS & DATA LANDSCAPE

Source systems feeding the integrated Northstar model

Source domain	Primary information
HRIS / Compensation	Employees, job history, pay, workforce snapshots, position bands
ATS	Recruiting requisitions and hiring activity
Timekeeping	Attendance, overtime, PTO
LMS	Training history and completion
CRM	Customer accounts, contacts, sales opportunities
Subscription / Billing	Plans, add-ons, subscription events, invoices, payments
Customer Success	Onboarding / implementation outcomes
Support	Tickets, SLA performance, escalation
Product Analytics	Monthly usage and adoption
PSA / Projects	Professional-services projects and time entries
Finance	Chart of accounts, cost centers, expenses, budgets, vendors

Target architecture



Raw source files remain evidence of the original system state. Cleanup and business rules occur in staging before trusted production tables are loaded.

Data-governance principle

Do not silently repair source data. Preserve the raw record, document the quality issue, define the transformation rule, and make the cleaned value traceable. The packet's data-quality register is the first example of this approach.

06 KPI & DECISION FRAMEWORK

What leadership needs the system to answer

Decision area	Representative KPIs
Growth	MRR • ARR • New ARR • Expansion ARR • Contraction ARR • Churned ARR • Win rate • Sales-cycle length
Customer	Logo churn • Revenue retention • Product adoption • Active-user % • CSAT • Support tickets per customer
Delivery	Time to go-live • Implementation on-time % • Project hours vs budget • Billable utilization • Escalation rate

Support

Ticket volume • SLA attainment • Resolution hours • Backlog • Priority mix • Repeat / escalated issues

People

Headcount • Turnover • Overtime • Absence • Training completion • Time to fill • Internal movement

Finance

Revenue mix • Gross margin • Operating income • Budget variance • A/R aging • Customer profitability • Project margin

Cross-functional questions the database should support

- Do customers with slower implementations show lower adoption, more support demand, or higher churn?
- Which plans or customer segments generate the most support workload relative to recurring revenue?
- Are expansion opportunities concentrated among customers with stronger adoption and CSAT?
- Which delivery teams or roles are experiencing overtime, workload, or staffing pressure as customer demand grows?
- Which professional-services projects are profitable after labor cost is considered?
- Are budget misses caused by revenue shortfall, cost growth, staffing, service inefficiency, or a planning assumption that proved wrong?

07 GOVERNANCE & BUSINESS-PLANNING ARTIFACTS**What should exist around the system**

Northstar does not need hundreds of artificial documents. A focused operating-system portfolio is enough to demonstrate how business planning, process design, governance, and data connect.

Artifact	Purpose	Northstar approach
Org chart	Reporting structure and span of responsibility	Build next / lightweight
Customer lifecycle map	End-to-end commercial and service journey	Defined in this model
Support escalation flow	Priority, SLA, escalation, closure	Defined in this model
RACI / responsibility matrix	Cross-functional ownership	Defined at high level
SOP library index	Controlled list of core procedures	Build later
Selected SOPs	Customer onboarding, support escalation, month-end reporting, data quality	Build later; AI can accelerate
Policy framework	Data access, change control, documentation, training/compliance	Build later; sample set only
Risk register	Operational, customer, people, vendor, security, financial risks	Build later
Succession / key-role continuity plan	Coverage for critical leadership and technical roles	Build later
Business continuity framework	System outage, vendor failure, staffing disruption	Build later
KPI dictionary	Metric definition, owner, source, calculation, cadence	Build as SQL/BI develops

Scope control

These supporting artifacts should demonstrate business thinking, not become a second full-time project. The highest-value learning now is the database, SQL, analytics, and application layer; familiar documentation can be accelerated with AI and then reviewed for business judgment.

08 REQUIREMENTS FOR THE MARIA DB BUILD**What this operating model means for the technical design**

Preserve history: Subscriptions, job changes, pricing, ownership, churn, and financial activity must not be reduced to current-state fields only.

Separate source from trusted data: Raw imports belong in staging; normalized tables should contain validated, mapped, business-ready data.

Use stable keys: Employee, customer, subscription, project, ticket, invoice, product, and account identifiers should support reliable joins.

Model many-to-many relationships explicitly: Examples include employees working many projects and customers purchasing multiple add-ons/services.

Support time-based analysis: Monthly workforce, product usage, revenue, expenses, budgets, and subscription events must be queryable over time.

Protect business rules: Plan pricing, statuses, account types, cost centers, and required relationships should use constraints/reference tables where appropriate.

Make anomalies traceable: Bad source values should be quarantined or flagged, not silently overwritten.

Design for reporting: The schema should make P&L, BvA, SaaS metrics, support performance, workforce analysis, and customer profitability achievable without spreadsheet workarounds.

Immediate next phase



The next work should shift from business planning into MariaDB architecture and hands-on database development.

Northstar is a synthetic environment. The value of the project is not the fictional company itself; it is the evidence of how business requirements, processes, systems, data, controls, and decisions are connected.